Unit II

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2.1 The need for layered architecture (protocol hierarchies):

What is Layered Architecture?

Layered architecture, also known as **protocol hierarchy**, is a design approach where the communication process is divided into **separate layers**, each with a specific function. A well-known example is the **OSI (Open Systems Interconnection) model**.

Need for Layered Architecture:

1. Simplifies Network Design:

Dividing the network into layers makes it easier to design, develop, and understand. Each layer performs a specific task.

2. Modularity:

Each layer is independent, so changes in one layer do not affect others. This makes **maintenance and upgrades** easier.

3. Interoperability:

Standard layers allow **different hardware and software** from various vendors to communicate effectively.

4. Troubleshooting and Debugging:

Problems can be **isolated layer by layer**, which simplifies error detection and resolution.

5. Flexibility and Scalability:

New protocols or technologies can be added in specific layers without redesigning the whole system.

In summary, layered architecture is essential for organizing complex network communication into manageable, independent parts that ensure compatibility, scalability, and ease of maintenance.

2.2 OSI reference Model:

What is the OSI Model?

The **OSI (Open Systems Interconnection)** model is a **theoretical framework** developed by ISO (International Organization for Standardization) that defines **how data is transmitted** over a network in **7 distinct layers**. Each layer performs a specific function and communicates with the layers directly above and below it.

7 Layers of OSI Model:

1. Physical Layer:

Deals with **transmission of raw bits** over physical medium (cables, switches).
Example: Ethernet cables, connectors.

2. Data Link Layer:

Responsible for **node-to-node delivery**, **error detection**, and **MAC addressing**.
Example: Switches, MAC address.

3. Network Layer:

Handles **routing** and **logical addressing (IP)** to deliver data across networks.
Example: Routers, IP address.

4. Transport Layer:

Ensures **reliable data transfer**, **flow control**, and **error recovery**.
Example: TCP, UDP.

5. Session Layer:

Manages **sessions** or **connections** between applications.
Controls dialog management.

6. Presentation Layer:

Handles **data translation**, **encryption/decryption**, and **compression**.
Example: Encoding formats like JPEG, ASCII.

7. Application Layer:

Closest to the user; provides **network services** to applications.
Example: HTTP, FTP, SMTP.

In Summary:

The OSI model helps in **standardizing network communication**, making it **easier to develop interoperable hardware/software**, and to **troubleshoot network issues** layer by layer.

2.3 TCP/IP reference Model:

What is the TCP/IP Model?

The **TCP/IP (Transmission Control Protocol/Internet Protocol)** model is a **conceptual framework** used to describe how data is transmitted over the Internet. It is the foundation of the modern **Internet** and has **4 layers**.

Layers of TCP/IP Model:

1. Application Layer:

Provides **network services directly to users and applications**.
Supports protocols like **HTTP, FTP, SMTP, DNS**.
Equivalent to the top three layers of the OSI model (Application, Presentation, Session).

2. Transport Layer:

Responsible for **end-to-end communication**, **error checking**, and **flow control**.
Main protocols:

TCP (Transmission Control Protocol) – reliable connection.

UDP (User Datagram Protocol)** – faster but less reliable.

3. Internet Layer:

Handles **logical addressing (IP addresses)** and **routing of data packets** across networks.
Key protocols: **IP, ICMP, ARP**.

4. Network Access Layer (Link Layer):

Deals with **physical transmission of data** over a network medium.
Includes hardware addressing (MAC), device drivers, and network interface cards (NICs).
Equivalent to the OSI's **Data Link** and **Physical** layers.

In Summary:

The TCP/IP model is a **simplified and practical model** used in real-world networking. It ensures **efficient communication over the Internet**, and serves as the **standard protocol suite** for all internet-based communication.

2.4 Ethernet Technology - Types of Ethernet, properties of Ethernet, Collision detection and Recovery, Ethernet hardware address, Ethernet Frame Format:

1. Types of Ethernet:

Standard Ethernet (10 Mbps) – Also known as **10Base-T**.
Fast Ethernet (100 Mbps) – Known as **100Base-TX**.
Gigabit Ethernet (1 Gbps) – Known as **1000Base-T**.
10-Gigabit Ethernet (10 Gbps) – Used in high-speed backbone networks.

Each type differs in **speed**, **cabling type**, and **usage area**.

2. Properties of Ethernet:

Uses CSMA/CD for managing access to the network.
Broadcast network – All nodes share the same communication channel.
Star topology with switches is common.
Supports scalable data rates from 10 Mbps to 100 Gbps.
Based on the **IEEE 802.3** standard.

3. Collision Detection and Recovery (CSMA/CD):

Carrier Sense Multiple Access with Collision Detection (CSMA/CD) is used.
Before sending data, a device **listens to the channel**.
If two devices transmit at the same time, a **collision** occurs.
Devices detect the collision and send a **jam signal**.
Then, they wait a **random backoff time** before retransmitting.

> Note: CSMA/CD is mainly used in **older half-duplex Ethernet**. Modern Ethernet with switches uses **full duplex** and avoids collisions.

4. Ethernet Hardware Address (MAC Address):

Each Ethernet device has a unique **48-bit MAC (Media Access Control) address**.
Example: `00:1A:2B:3C:4D:5E`.

Used to identify devices at the **Data Link Layer**.

5. Ethernet Frame Format:

Preamble	Destination MAC	Source MAC	Type/Length	Data	CRC

Preamble (7 bytes): Synchronization bits.
Start Frame Delimiter (1 byte): Marks start of frame.
Destination MAC (6 bytes): Receiver's address.
Source MAC (6 bytes): Sender's address.
Type/Length (2 bytes): Indicates payload type.
Data (46–1500 bytes): Actual data.
CRC (4 bytes): Error checking (Cyclic Redundancy Check).

In Summary:

Ethernet is the most widely used LAN technology. It supports various speeds, uses MAC addresses for device identification, manages collisions using CSMA/CD (in older systems), and transmits data using a standardized frame format.

2.5 Wireless LAN:

What is Wireless LAN?

A **Wireless LAN (WLAN)** is a **local area network** that uses **wireless communication (radio waves)** instead of physical cables to connect devices like laptops, smartphones, and printers within a limited area such as a home, office, or campus.

Key Features of WLAN:

1. Uses IEEE 802.11 Standards:

Commonly known as **Wi-Fi**.
Supports various versions like 802.11a/b/g/n/ac/ax.

2. Wireless Access Point (WAP):

Acts as a hub for connecting wireless devices to a wired network or the Internet.

3. SSID (Service Set Identifier):

Unique name used to identify a WLAN.

4. Security Mechanisms:

Includes **WEP, WPA, WPA2, and WPA3** for encrypting wireless communication.

Advantages of WLAN:

Mobility: Users can move freely within the coverage area.
Ease of Installation: No physical cabling is needed.
Scalability: Devices can be easily added to the network.
Cost-Effective: Saves on wiring and maintenance costs.

Disadvantages of WLAN:

Interference: Susceptible to radio interference and signal loss.
Security Risks: Wireless signals can be intercepted if not encrypted properly.
Limited Range: Coverage depends on the strength of the access point.

In Summary:

A Wireless LAN provides **flexible and mobile network access** using radio waves, making it ideal for modern environments that require easy connectivity and minimal wiring.

2.6 Bluetooth:

What is Bluetooth?

Bluetooth is a **wireless communication technology** used for **short-range data exchange** between electronic devices using **radio waves** in the **2.4 GHz ISM band**. It enables the creation of **personal area networks (PANs)**.

Key Features of Bluetooth:

1. Short Range Communication:

Typical range is **10 meters**, extendable up to **100 meters** with enhanced power.

2. Low Power Consumption:

Designed for devices like smartphones, headsets, keyboards, and smartwatches.

3. Device Pairing:

Devices must be **paired** to securely communicate with each other.

4. Operates in Piconets:

One master device can connect with up to **7 active slave devices**.

5. Versions:

Different versions like **Bluetooth 4.0 (Low Energy)**, **5.0**, **5.2**, offering improved **speed, range, and efficiency**.

Advantages of Bluetooth:

Wireless Convenience: No need for cables to connect devices.
Low Cost and Easy Setup: Inexpensive and simple to configure.
Widely Supported: Built into most modern smartphones, laptops, and accessories.
Secure Connection: Offers encryption and frequency hopping to reduce interference.

Disadvantages of Bluetooth:

Limited Range: Effective only over short distances.
Lower Data Speed: Slower than Wi-Fi or other wireless technologies.
Interference: Can face interference from other wireless devices in the same frequency band.

In Summary:

Bluetooth is a **reliable and low-power** wireless technology used for **short-range communication** between devices, making it ideal for **hands-free calling, audio streaming, file sharing**, and **IoT applications**.